Design and fabrication of a Solar Still with an Internal Condenser

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Abstract- Solar energy is a clean, abundant, and sustainable resource that plays a crucial role in addressing global water scarcity through desalination technologies. This study focuses on the development of a solar still that harnesses solar energy efficiently to produce freshwater. The system incorporates Phase Change Material (PCM) and an internal condenser to enhance heat retention, extend evaporation time, and improve condensation rates, ultimately increasing water yield. A parametric analysis is performed to evaluate the effect of different materials on thermal efficiency and system performance. Furthermore, the use of hydrophobic films and coatings minimizes droplet adhesion, facilitating better condensate collection. By optimizing material selection and thermal management, this research contributes to the development of high-performance, energy-efficient desalination systems, leveraging solar energy as a sustainable solution for freshwater generation.

Keywords- Solar Still, Internal Condenser, Desalination, single slope

I. Introduction

Access to safe drinking water remains a critical issue, with approximately two billion people worldwide lacking reliable access to clean water. In response to this challenge, the United Nations designated 1981–1990 as the International Drinking Water Supply and Sanitation Decade (IDWSSD). Approximately 60% of the global population faces freshwater scarcity. Consequently, the reliance on natural water sources has become essential; however, many of

these sources are heavily contaminated with household and industrial waste, rendering them unsuitable for consumption. Additionally, transporting potable water via trucking or extensive pipelines from distant locations is often economically unfeasible, posing significant logistical challenges for both human and livestock needs. Given that most of Earth's water is saline and unsuitable for direct consumption, finding an economically viable method for converting brackish water into fresh water could provide a partial solution to the water scarcity issue. Desalination, the process of transforming saline water into safe, usable water, offers an affordable means of securing drinking water. This technology can contribute to significant improvements in various aspects of life, including health and hygiene, livelihoods (both urban and rural), school attendance, psychological well-being, and social cohesion. Desalination relies on energy sources such as fossil fuels, electricity, or solar energy to separate saline or brackish water into fresh water. The global capacity for desalinating brackish water is expanding rapidly, driven by the increasing demand for clean water in growing populations. As reported by the International Desalination Association (IDA), there are currently 14,000 desalination plants operating worldwide, collectively processing 60 million cubic meters of water daily.

The performance of single-slope solar stills (SSSS) has been significantly improved through the integration of Phase Change Materials (PCMs) and auxiliary technologies. These enhancements address the limited operational hours of conventional solar stills (CSS), primarily restricted to daytime. One approach combined a solar air heater (SAH) with PCMs, which raised water temperatures and extended system operation into the evening. This improved freshwater output but introduced added cost and complexity, with performance influenced by environmental conditions [1]. Another study showed that using PCM alone as thermal storage extended desalination into nighttime hours. The PCM-based still produced 7.54 L/m²/day—67.18% more than a conventional still—and also slightly reduced the cost of water production, showing promise for arid regions [2]. Further improvements were achieved by enhancing PCMs with nanoparticles to form nano-PCMs (NPCMs). Adding aluminum oxide to paraffin wax increased thermal conductivity, boosting productivity by 60.37% and efficiency by 68.29%, compared to a conventional system [3]. Paraffin wax remains a preferred PCM, especially when enhanced with additives like copper oxide. This combination can raise productivity by up to 180% and efficiency by 67.2%, though challenges like low base conductivity and higher system costs persist [4].

The inclination angle of a solar still's glass cover plays a crucial role in optimizing solar radiation capture and condensation efficiency. Several studies have explored this parameter across different regions and configurations to determine the ideal angle for maximum productivity. A.A. Azooz examined ten solar stills with inclination angles ranging from 10° to 55° in 5° increments. The findings showed that angles between 20° and 25° provided the best performance, while angles between 30° and 35° were associated with a noticeable drop in efficiency, highlighting the importance of precise angle selection [5]. Vikas Kumar Thakur conducted experiments with glass angles set at 11°, 26°, and 41°, corresponding to 15° below, equal to, and 15° above the latitude of Gwalior, India (26.2183°N). The highest freshwater yield was achieved at 26°, aligning with the local latitude. This emphasizes the advantage of setting the glass angle equal to the geographic latitude for optimal performance [6]. Y.A.F. El-Samadony tested a stepped solar still using glass angles of 25°, 27.5°, 30°, 32.5°, and

35°, along with flat and convex absorber plates. The optimal configuration combined a 32.5° glass angle with a convex absorber plate, maximizing evaporation and condensation efficiency [7]. In a simulation-based study, Ayad Atiyah used SOLIDWORKS and ANSYS CFD to model solar still performance for Baghdad's climate. Analysis conducted on July 25, 2023, identified 32° as the optimal glass cover inclination angle using a custom MATLAB algorithm. This confirmed the importance of climate-specific design optimization [8].

The efficiency of solar stills has been further explored by modifying their geometry and incorporating thermal enhancement techniques. Among these, comparisons between single-slope and double-slope configurations have offered insights into design optimization. One study examined the use of pin fin absorber plates in both single- and double-slope solar stills. At a consistent water depth of 3 cm, the double-slope design yielded 4.4 kg/m²·day, outperforming the single-slope at 3.6 kg/m²·day. Pin fins enhanced heat transfer, improving evaporation and condensation rates. However, these gains came at the expense of increased material costs and system complexity [9]. Another investigation evaluated the performance of single-, double-slope, and pyramidal solar stills across various orientations and tilt angles. The single-slope still, under optimal alignment, achieved a 28% increase in productivity. Yet, its efficiency was influenced by seasonal and geographic conditions, limiting its consistency across different environments [10]. A further study introduced an elevated basin into a double-slope solar still to reduce thermal losses. This design enhancement led to a 36.7% increase in freshwater output. Despite the improved performance, the approach involved greater fabrication complexity and higher initial investment [11]. Finally, enhancements using biomass and energy-storing materials were tested on both single- and double-slope stills. The single-slope variant demonstrated superior productivity, especially when biomass was used to assist heating. Nevertheless, the reliance on a continuous biomass supply and additional maintenance posed operational challenges [12].

Recent developments in solar still technology have explored innovation to enhance performance. Rajaseenivasan et al. [13] integrated a vacuum fan with a single slope solar still, facilitating faster vapor removal and increasing the daily yield by 45%. Elashmawy [14] optimized the thermal insulation of basin liners using jute cloth and reflective coatings, achieving a 32% rise in freshwater productivity due to reduced thermal losses. Bait and Si-Ameur [15] introduced a solar still design incorporating rotating wick materials to continuously renew the evaporative surface, resulting in an approximately 40% enhancement in output. Meanwhile, Yadav and Agrawal [16] experimented with the addition of multiple glass covers and cooling techniques to improve condensation rates, leading to a 30–35% increase in daily water production compared to conventional designs. These studies collectively emphasize the potential of mechanical enhancement, surface renewal, and better thermal management in advancing solar still efficiency. Recent advancements in surface modification techniques have played a key role in enhancing the efficiency and output of solar still desalination systems. These studies focus on improving condensation and evaporation processes through plasma treatments and nanocoatings. One study applied cold plasma coating using Dielectric Barrier Discharge (DBD) to enhance surface hydrophobicity in solar stills. The modified surface increased freshwater output by 25.7%, from 485 ml to 610 ml, under 1000 W/m² solar radiation, with an absorber wall temperature of 70.1°C and a 35° surface angle. The distilled

water met WHO standards, indicating the viability of plasma treatments for improved performance [17]. In another investigation, silicone nano coatings were applied to the glass condensation surface of solar stills. This modification enhanced surface wettability, leading to a 20% increase in freshwater production, thereby demonstrating the effectiveness of condensation surface treatment in boosting productivity [18]. A related study examined the impact of varying the coated surface area using a commercial water-based silicone solution. A 30% coated area yielded the highest freshwater output, while coating beyond 50% increased thermal resistance and cover temperature. Coatings below 50% proved more cost-effective than uncoated glass, emphasizing the benefits of mixed wettability surfaces for optimal performance [19]. Further improvement was achieved through oxygen plasma treatment and graphene-enhanced surfaces, which improved water adhesion and salt rejection. This approach led to a 48.9% rise in water recovery—from 2.90 to 4.32 L/m² per day. It also significantly improved water quality, with a 96.52% reduction in conductivity and salinity, and a notable pH increase. These modifications enhanced both efficiency and output purity [20].

Water depth has a significant impact on solar still performance, as it directly influences the rate of water heating and evaporation. Several studies have examined optimal depths across different still designs and configurations to maximize freshwater yield. Phadatare and Verma (2007) investigated the performance of a plastic solar still and found that a 2 cm water depth produced the highest yield at 2.1 L/m²/day. Deeper water layers reduced efficiency due to slower heating. They also noted the benefits of Plexiglas in resisting corrosion and emphasized that evaporative heat transfer was the primary mechanism driving productivity [21]. Balamurugan et al. (2017) compared passive solar stills with those coupled to flat plate and evacuated tube collectors. Their findings confirmed that a shallow depth of 1 cm maximized output. Among the designs, stills with evacuated tube collectors performed best due to superior water preheating [22]. Younis et al. (2010) explored how water depth, salinity, and glass cover thickness affect distillation efficiency. They found that a lower water depth of 6 cm, along with reduced salinity (28 mmoh/cm) and a thicker glass cover (6 mm), resulted in the highest yield of 5.1 L/m²/day. These parameters contributed to better thermal management and evaporation [23]. Muthu Manokar et al. (2019) evaluated an acrylic pyramid solar still and found that a 1 cm depth yielded the most freshwater—3.72 kg/m²/day with insulation and 3.27 kg/m²/day without. Increased water depth reduced productivity, as it slowed down water heating and evaporation rates [24].

Another enhancement involved attaching an external condenser to a solar still as described in [25], where condensation occurred not only on the glass surface but also along four sidewalls cooled via circulating water tubes. This setup contributed to improved efficiency through enhanced condensation. Nijegorodov et al. [26] explored the use of a passive condenser in conjunction with a single slope solar still. In their approach, saturated air and water vapor were extracted from the still using a low-power exhaust fan and passed through a condenser. This modification resulted in a more than twofold increase in thermal efficiency and nearly doubled the distilled water yield. Rahim [27] introduced a forced condensing method within a moving film inclined solar desalination system. This design included four key components and involved extracting vapor before it reached the glass cover. The vapor was condensed in a separate, cooler unit. The system achieved thermal and overall

efficiencies of approximately 70% and 60%, respectively. Kabeel et al. [28] proposed another modification for a single basin solar still to boost distilled water output. Their setup included a saline water tank, a traditional still, and a modified basin still integrated with an external condenser, utilizing a vacuum fan. Their experiments showed that the external condenser increased distillate yield by around 53.2%. Additionally, mixing aluminum oxide nanoparticles with the basin water—creating a nanofluid—further enhanced productivity. When combined with the external condenser, this nanofluid approach increased water output by about 116%.

Research efforts have focused on enhancing the productivity, efficiency, and yield of solar stills by integrating internal condensers. In a relevant experimental study, Ahmed investigated the performance of a single-effect solar still equipped with an internal condenser. The findings demonstrated that this configuration improved water productivity by approximately 30% compared to a conventional solar still [29]

As there has been very little work done on internal condenser in recent years especially in the context of India this study chose to study the effects of an internal condenser in the climatic conditions of India.

II. Materials and Methods

1. Absorber plate:

The absorber plate and overall structure of the solar still were fabricated using Galvanized Iron (GI) sheet with a thickness of 0.5 mm. GI was chosen due to its balanced combination of affordability, good thermal conductivity, and corrosion resistance compared to other commonly used materials such as Stainless Steel (SS) and Mild Steel (MS). While SS is highly resistant to corrosion, it is considerably more expensive. On the other hand, MS, although cheaper, tends to rust over time, especially in high-humidity environments, making it unsuitable for this application. To enhance the thermal absorption of the still and to reduce the problem of scaling, the inner surface of the GI structure was coated with black epoxy paint. This type of paint not only improves the absorption of solar radiation due to its high emissivity but also serves as a protective layer that reduces the risk of chemical reactions between the metal and water. Three coats of black epoxy paint were applied to the internal surfaces, with each coat allowed to dry overnight for proper curing and durability. This measure significantly helped in maintaining higher internal temperatures and prolonged the operational lifespan of the system. The structural design also incorporated specific slots at the top edges of the GI body to securely fit the glass cover at a fixed inclined angle. This design ensured proper sealing, efficient condensation flow, and ease of glass removal for maintenance. Additionally, an access hole was included in the body of the still to facilitate cleaning and regular maintenance. This hole allowed for the removal of salt and mineral deposits that accumulate over time, ensuring smooth operation and hygienic water output. To collect the condensed water, a V-shaped duct was fabricated from the same GI material and installed along the lower edge of the glass cover. This ensured material uniformity and prevented compatibility issues that might arise from using dissimilar materials. The GI duct provided a durable, rust-resistant pathway for the distilled water, channeling it efficiently into the collection container with minimal loss or contamination.



Figure 2. Main Body of still welded with 20 gauge GI

2. Insulating Material



Figure 3. Expanded Polyethylene

In order to minimize thermal losses and improve the energy efficiency of the solar still, Expanded Polyethylene (EPE) was used as the primary insulating material. Insulation is a critical component in passive solar distillation systems, as it helps in retaining the absorbed solar energy within the system, thereby maintaining elevated water temperatures for longer durations and enhancing the rate of evaporation.EPE was selected based on its advantageous physical and thermal properties. It exhibits a relatively low thermal conductivity of 0.3 W/m·K, which makes it an effective barrier against conductive heat losses. Additionally, its lightweight structure, mechanical resilience, and resistance to moisture make it a durable and efficient insulation material for outdoor applications. In contrast to materials such as thermocol or glass wool, EPE offers superior flexibility and impact resistance, making it particularly suitable for systems subjected to periodic handling and environmental exposure.The EPE sheets used in this study were pre-laminated with a factory-applied waterproof foam layer, which eliminated the need for additional external waterproofing. This built-in protective layer ensured resistance to moisture ingress and extended the operational life of the insulation without compromising thermal performance.The sheets were custom-cut and externally applied to the bottom and vertical sides of the galvanized iron (GI) still using adhesive bonding. This configuration provided a continuous insulation envelope around the basin area, effectively reducing heat transfer from the metal surface to the ambient environment.The application of EPE insulation played a significant role in enhancing the thermal retention capability of the solar still, which, in turn, positively influenced the evaporation rate and the overall distillate yield. The integration of durable and thermally resistant insulation materials such as EPE contributes substantially to the long-term

efficiency and practicality of passive solar desalination systems.

3. Glass

The transparent top cover of the solar still was fabricated using a 5 mm thick single-pane glass sheet, which served the dual purpose of allowing solar radiation to enter the system and enabling the condensation of water vapor on its inner surface. The use of glass was preferred due to its high solar transmittance, durability, and ability to act as an effective condensing surface under varying ambient conditions.The glass sheet was mounted securely into pre-cut slots along the upper edge of the galvanized iron (GI) structure. It was inclined at an angle of 25°, which was chosen based on the geographic latitude of the experimental site to ensure optimal solar radiation capture and efficient drainage of condensate. This inclination also helped prevent the re-evaporation of condensed droplets and supported their smooth flow into the collection duct by gravity. The selected 5 mm thickness provided sufficient structural strength to withstand external mechanical and thermal stresses while maintaining good optical clarity. The smooth inner surface of the glass promoted effective condensation of the water vapor rising from the heated basin. The condensed droplets flowed downward along the inclined glass surface and were collected at the lower edge into a V-shaped galvanized iron duct.The integration of a well-supported and inclined glass cover played a crucial role in enhancing both the energy absorption and the water collection efficiency of the solar still. It contributed significantly to the overall thermal performance and operational reliability of the system.

4. Internal condenser:

An internal condenser made of copper was installed on the inner side of the back plate of the solar still. The condenser consisted of pipes with a 0.5-inch diameter and 9 horizontal tubes. Its inlet and outlet were connected to a water source at an elevated position, ensuring continuous water circulation. The condenser facilitates the condensation of water droplets on its surface, allowing addittional collection from the outlet, thereby increasing the output and efficiency of the still.

A. Design of internal condenser:

The design of internal condenser was carried out accounting for manufacturability transportation assembly and space constraints theoretical analysis was carried out to find the optimum length as the theoretical analysis requires assumption of either diameter or the length the diameter was assumed to be 12mm the main problem to tackle was the changing flow regime and the changing ambient temperature throughout the day hence the internal condenser was designed for the peak temperatures in summer season.

Following equations were used to calculate the optimum length:

$$\mathrm{Re}_D = \frac{4\dot{m}}{\pi D \nu} \tag{1}$$

As the mass flow rate is very less , the flow can be assumed to be laminar , this results in in a lower value of h and a need for a longer condenser

$$h_{horiz} = 0.729\left[\frac{g\rho_l(\rho_l-\rho_v)h_{fg}^3 k_l^3}{\mu_l(T_{sat}-T_s)D}\right]^{1/4} \quad (\mathrm{W/m^{2}\,.^{o}C}) \tag{2}$$

$$\overline{h_o} = \frac{21338.77}{(100-T_w)^{1/4}}\ \mathrm{W/m^2 \cdot K}] \tag{3}$$

The wall temperature was assumed to be 90 degree Celcius and the outlet temperature was measured to be 70 degree Celcius

$$\overline{h_o}\pi D_o L(100 - T_w) = h_i \pi D_i L \left(\frac{T_w - 30 - (T_w - 70)}{\ln\left(\frac{T_w - 30}{T_w - 70}\right)} \right) \quad (4)$$

$$\overline{h_o}\pi D_o L(100 - T_w) = h_i \pi D_i L \left(\frac{70 - 30}{\ln\left(\frac{T_w - 30}{T_w - 70}\right)} \right)] \quad (5)$$

The above type of equations are solved using trial and error methods in textbooks or some kind of numerical analysis scheme like Newton-Raphson.

Solving for L gives the value to be around 7.5 m.

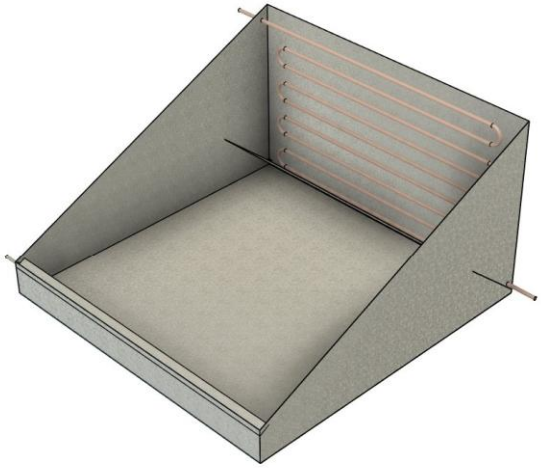


Figure 4. Final cad rendering of the assembly

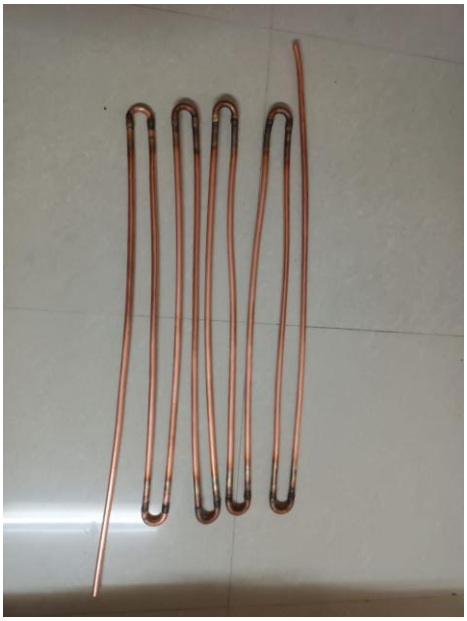


Figure 5. Final manufactured condenser

5) Output: For collecting the water droplets condensed on the internal condenser , a pvc pipe was cut in half and attached below it . It was given a slope so as condensed output water will slide down into the outside barrel. A GI frame pipe was attached on the front side of the still to collect the primary output water which condenses on the glass. to speed up water collection a diagonal GI strip was placed into the pipe so that the collected water will slide into the output barrel as quickly as possible , to avoid it being evaporated again .



Figure 6. Final Assembly

6) Data logger

Sr. No	Component	Range	Specification
1	T-type Thermocouple	0-150 (°C)	7/36 Swg., Teflon, 2 cores
2	Pyranometer	0-1000 (W/m2)	Silicon based, 0-5V input, 5-meter cable
3	Paperless Recorder	-	Unirec CZ (RS485), 230V AC, USB type
4	DAQ Card **Data Logger**	8 channels	Universal input type

Table 1. Data logger specifications

7) PCM

Analytical modelling :

QPCM is the energy stored within or released from PCM during day and night time, respectively, which can be expressed as:

$$Q_{PCM} = \frac{T_w - T_{PCM}}{R_{tot}} \times A_{PCM}, \quad \text{for } T_{PCM} \neq T_m \tag{6}$$

$$Q_{PCM} = \frac{m_{PCM} \times \lambda_{PCM}}{\Delta t}, \quad \text{for } T_{PCM} = T_m \tag{7}$$

mPCM = 0.325 kg

λ_{PCM} *= 226 KJ/kg*

where TPCM is PCM temperature, λPCM is the latent heat of PCM, mpcm is the mass of PCM (kg), Δt is the period of time. RTot is the total thermal resistance which can be expressed as:

$$R_{tot} = \frac{1}{h_w} + \frac{x_{tube}}{k_{tube}} \tag{8}$$

where xtube is the tube thickness containing PCM (taken to be 1 mm), ktube is the thermal conductivity of the GI basin (41 W/m K), hw is the convective heat transfer coefficient of water.

III. Results and Discussions

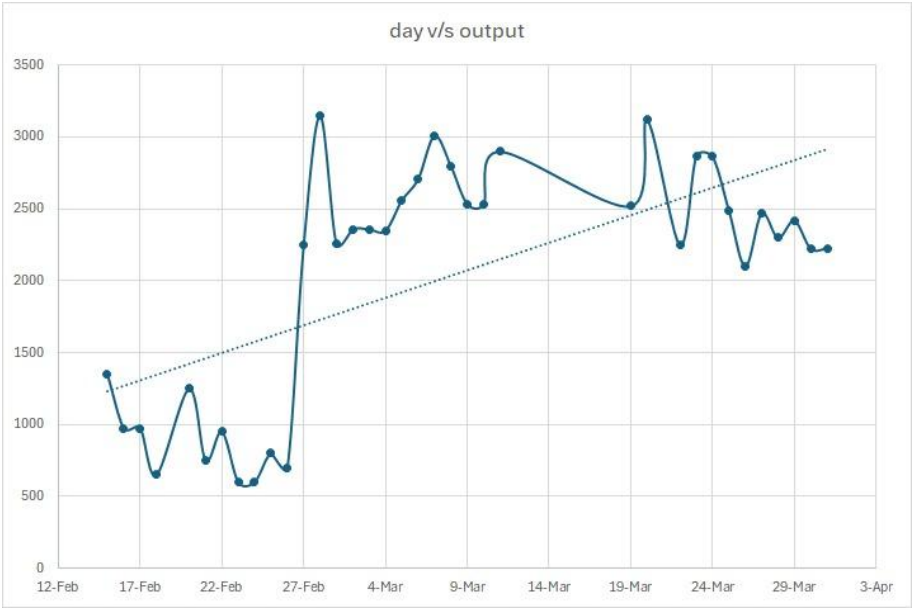


Figure 7. Output data of the still

A. Energy efficiency of the solar still :

$$Q_{\text{useful}} = m_d(c_w \Delta T + L_v) \quad (9)$$

$$\eta = \frac{m_d(c_w \Delta T + L_v)}{Q_{\text{incident}}} \quad (10)$$

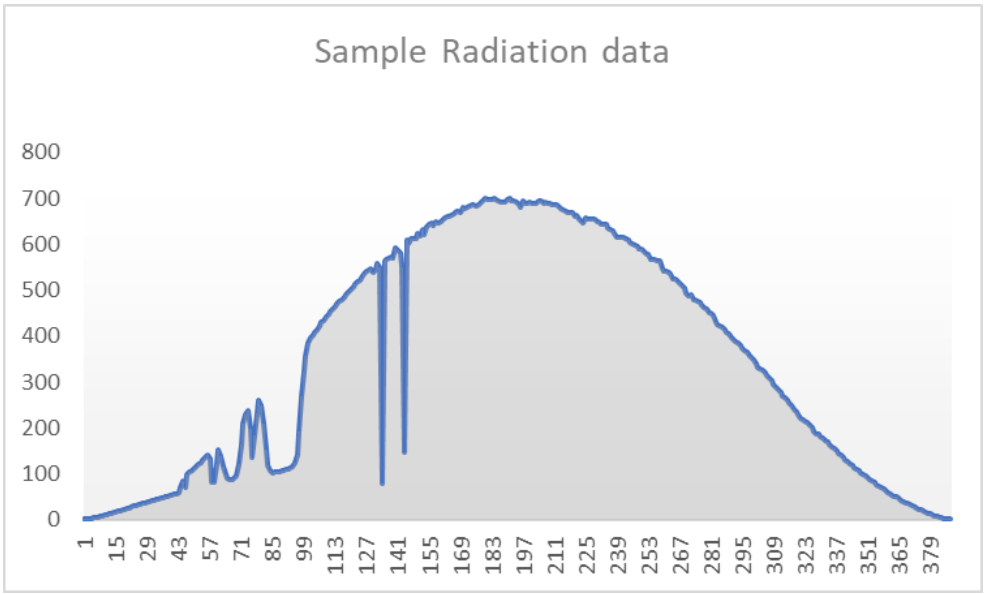


Figure 8. Area under this curve was calculated using Trapezoidal rule

$$\int_a^b f(x)\,dx \approx \frac{1}{2}h\left[y_0 + y_n + 2(y_1 + y_2 + \ldots + y_{n-1})\right]$$

$$\text{where } h = \frac{b-a}{n} \text{ and } y_r = f(x_r)$$

... (10)

Q incident is calculated by applying the trapezoidal rule on the radiation data , the formula was tested on various samples of radiation from , sunny day to a extremely cloudy day and the results were consistent , and efficiency of the solar still comes out to be 34%..

B. Internal condenser experiment :

An Experiment was carried out on the internal condenser to find the optimum mass flow rate the tap on the overhead tank can create a maximum of 4 LPM mass flow rate seven points of data will taken on 4 LPM 3 LPM 2LPM 1 LPM half LPM .3 LPM and 0.1 LPM ,

The condenser was kept open for half an hour.

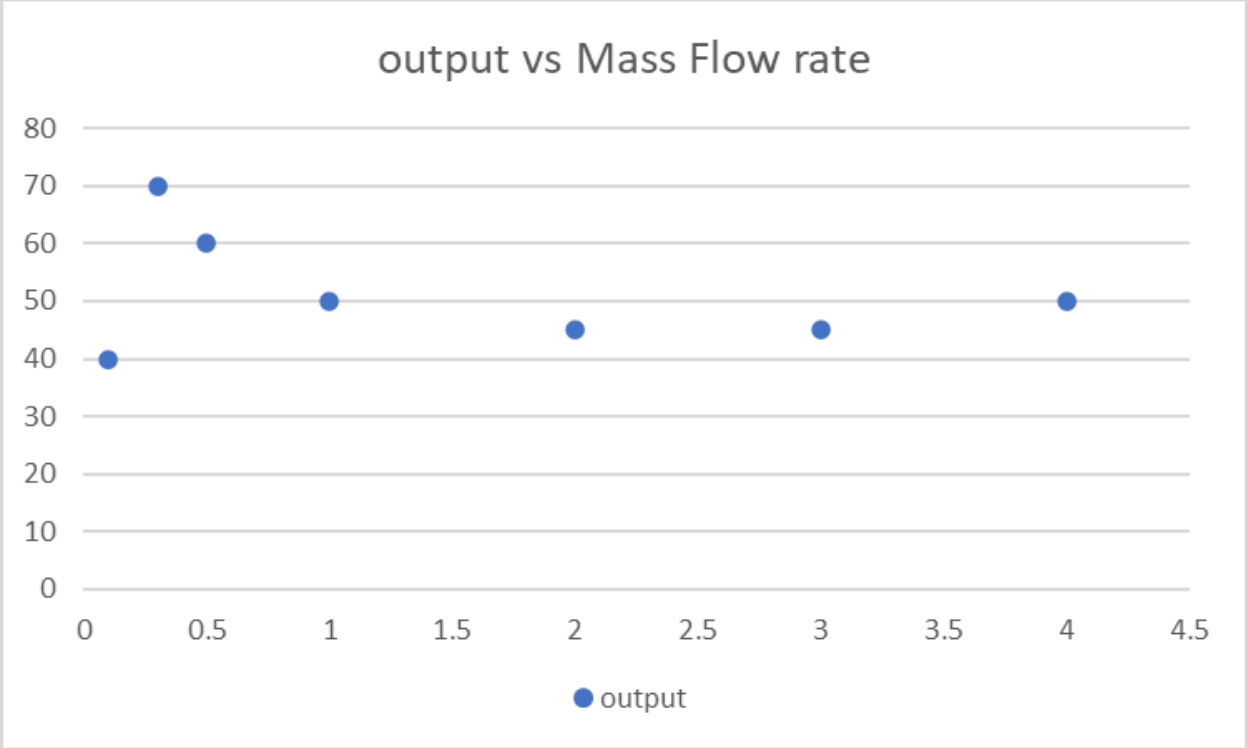


Figure 9. Results of the experiment on internal condenser

Though by logic it seems that maximum mass flow rate will result in the maximum output but maximum mass flow rate results in rapid saturation of all the vapor inside the steel, this results in emptying of all the vapor inside the still within two to three minutes, in our experiment 0.3 lpm was found out to be the optimum mass flow rate as it coincides with the vapor generation rate inside the still giving a slow but steady output in the half hour.

C. Water depth experiment :

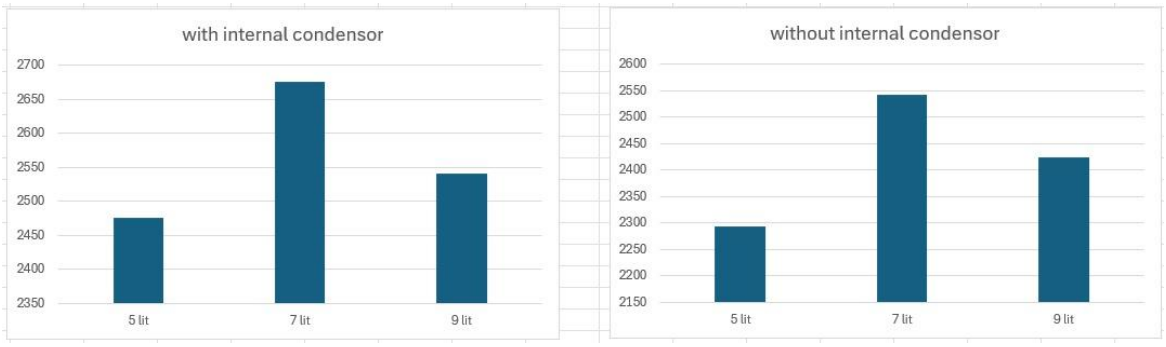


Figure 10. Optimum Water depth with condenser and without condenser

D. PCM experiment:

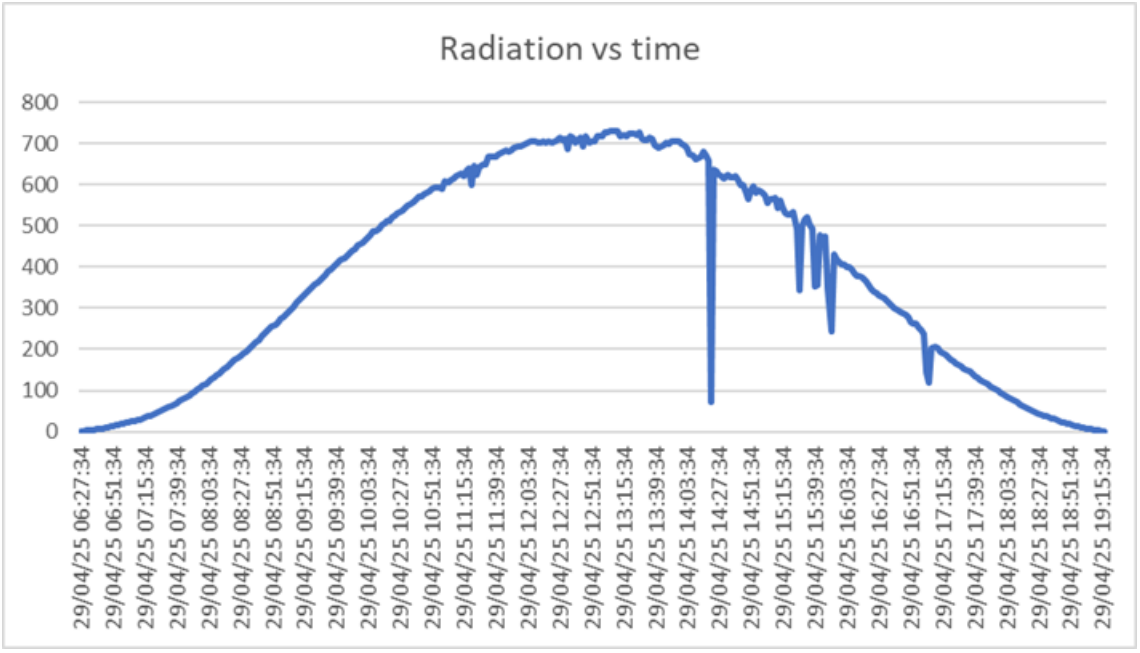


Figure 11 .Radiation data for 29th April

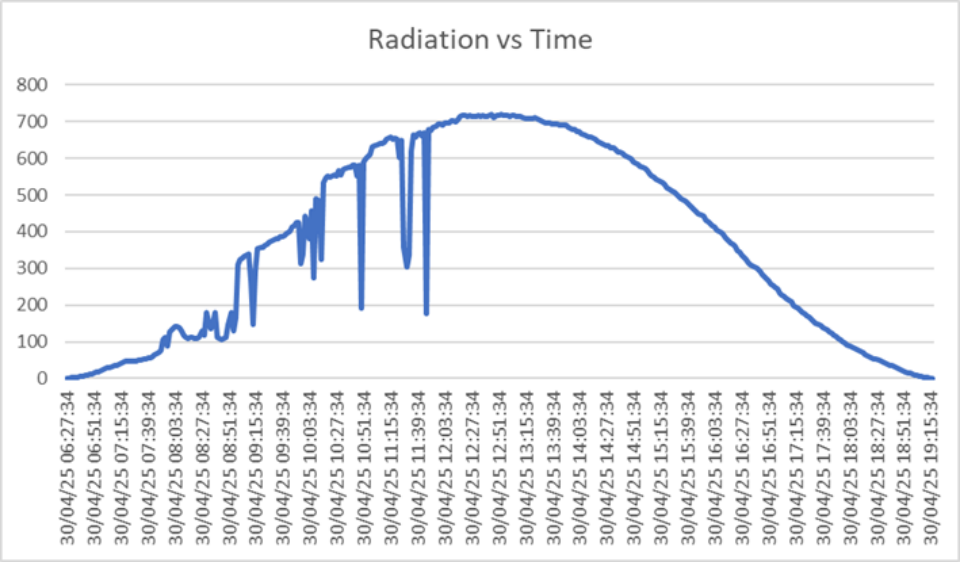


Figure 12. Radiation data for 30th April

First and foremost to make the distinction between the results one has to make sure that the obtained results are not due to the variation in radiation of the sun ,so the radiation profile of the whole day was plotted the key point to look at is the maximum radiation reached in the day in both the cases it is nearly 715 watts per meter square not only that the average radiation are also nearly equal, on 29th it is 377 and on the 30th it is 370.The experiment would have been hindered if the average radiation for the second day was more, as the average radiation only dropped any addition in the output would be result of the experimentation.

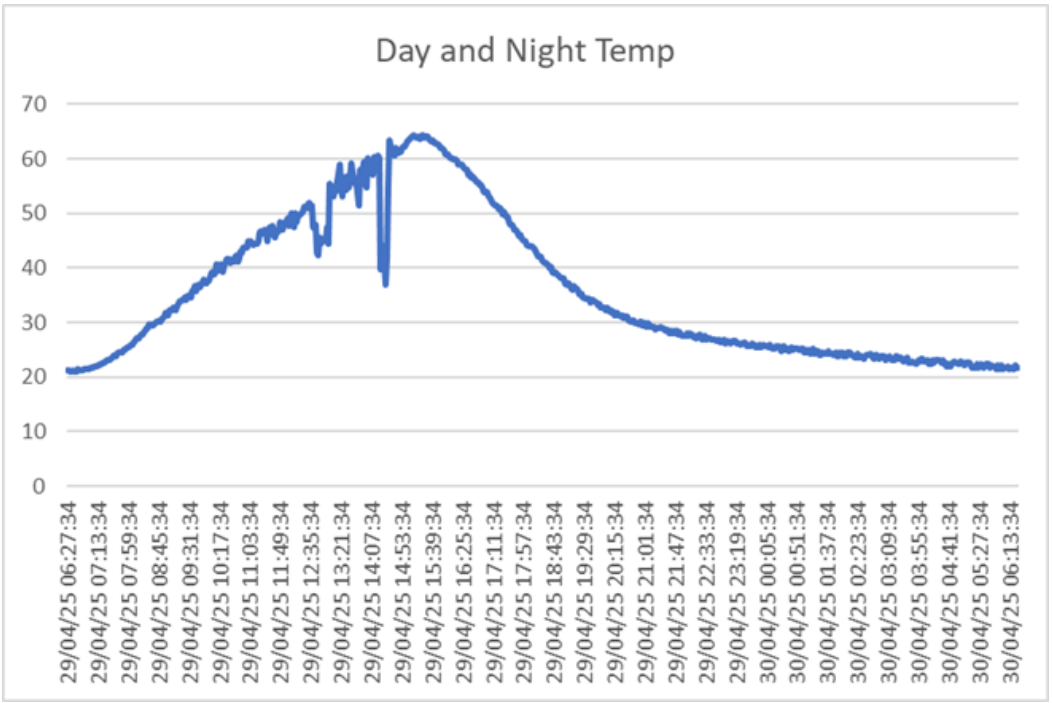


Figure 13. water temperature inside the still without addition of PCM

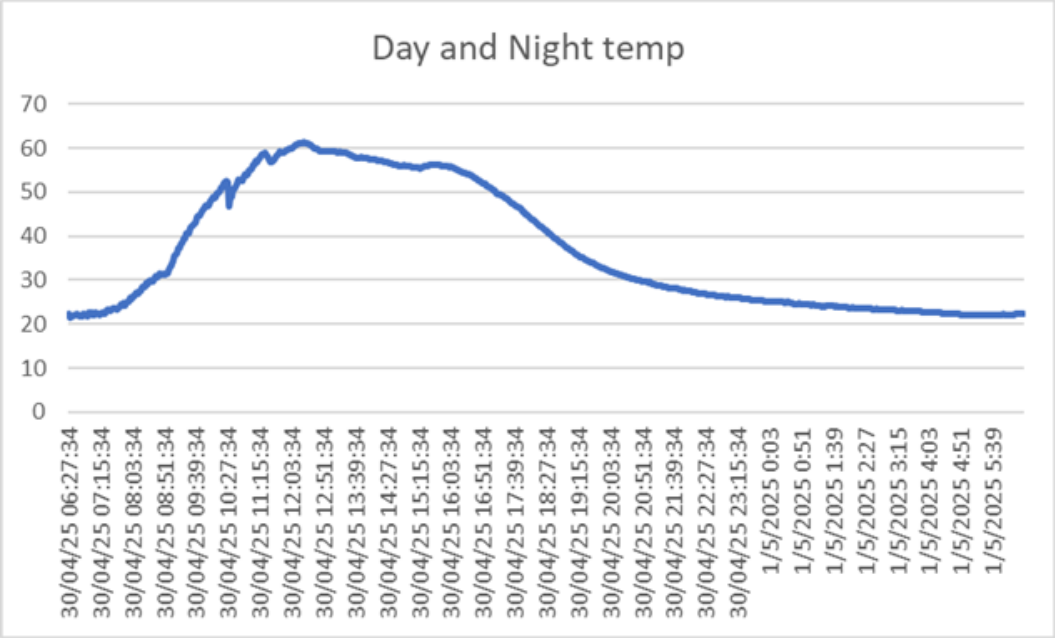
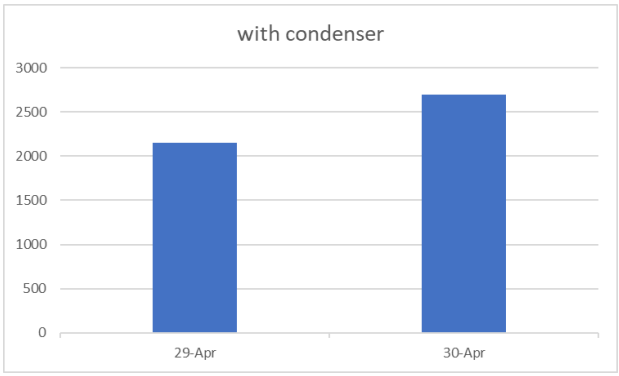
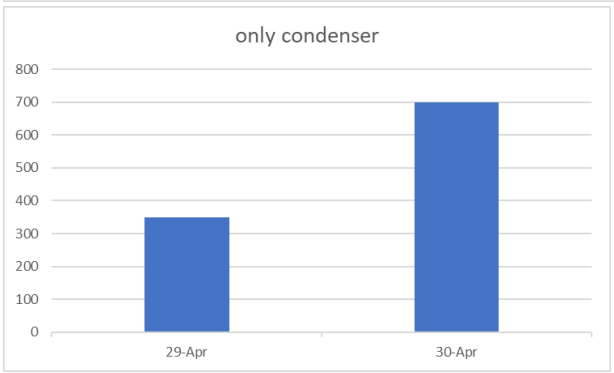


Figure 14. Temperature inside the still with PCM

The above temperature profile tells us about the effect PCM on the inside temperature of the solar still when the pcm was not added the water reached its peak at 3:00 PM as the sun's radiation gradually increased the temperature of water inside the still on the other hand in the next graph the temperature of water was increased rapidly due to addition of copper rods and was sustained for a longer time due to effect of giving out latent heat of the pcm material this benefited the internal condenser as a lot of vapor was available for the internal condenser to convert into condensed water.





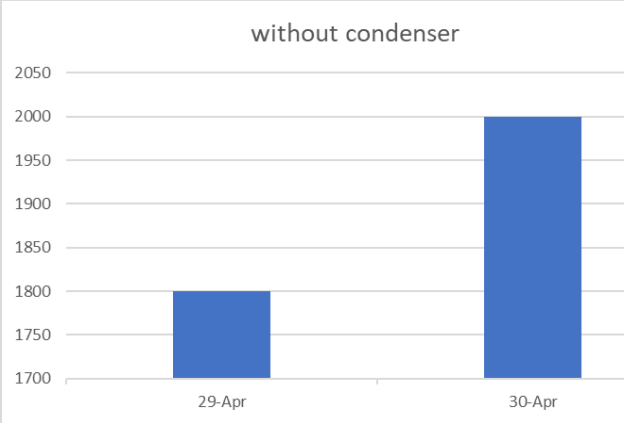


Figure 15. Comparative results of the PCM experiment

IV. Conclusion

Following are the conclusions we have arrived upon the analysis of the results obtained from the experimentation:

1. Galvanized iron along with 25 degrees still angle is the optimum configuration for a solar still installed in India for cost effectiveness and efficiency.
2. Though internal condenser and addition of PCM are studied extensively in previous research works our research highlights the additional gains when the two techniques are combined. Final results showed that output of the experiment there was a 25% increase in the net yield including the yield of the condenser.
3. The output of the condenser itself was doubled even though the radiation was same this very well showcases the effect of pcm on increasing the yield of the internal condenser.
4. Through the water depth experiment we realized that seven liters was the optimum volume of liquid to balance temperature inside the steel And maintain a continuous supply of vapor inside the still.
5. Through our interview condenser experiment we conclude that 0.3 LPM is the optimum mass flow rate which signifies importance of a slow and steady mass flow rate to coincide with the vapor generation rate inside the still.
6. Instead of calculating the efficiency of a solar steel using the water distillate data one must really look at the energy efficiency, that is how much percentage of energy received from the sun is actually converted into the distillate so as to compare different types of stills. Our still's energy efficiency is 34%.

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